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Problem statement: 52 - Community Tool & Equipment Library

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall calculate required security deposits based on tool category, enforce loan duration limits, and calculate late return fees.	**High**	**Pass:** Deposit, maximum loan duration, and late fee are correctly calculated according to the configured rules. **Fail:** Any of the three values is missing or incorrect.	Core requirement for controlling equipment loans and financial liabilities.
FR-002	Functional	The system shall allow community members to search for equipment and view its availability, category, and location.	**High**	**Pass:** A search for an equipment category displays all matching equipment with its current availability and depot location.	Enables members to find and select equipment before borrowing it.

FR-003	Functional	The system shall allow a community member to reserve available equipment for a specified borrowing period.	**High**	**Pass:** When an available item is reserved, the system records the member, equipment, reservation period, and changes its status to reserved. **Fail:** An unavailable item can be reserved.	Prevents conflicts between members requesting the same equipment.
FR-004	Functional	The system shall allow the library custodian to process equipment check-outs and returns while recording the associated dates and equipment status.	**High**	**Pass:** After checkout, the equipment is marked as borrowed with a recorded return date. After return, its status changes to returned/available or requires inspection.	Provides the core mechanism for tracking equipment throughout its loan lifecycle.

FR-005	Functional	The system shall allow the library custodian to record damage assessments for returned equipment and associate applicable damage charges with the member.	**Medium**	**Pass:** A custodian can record the damaged equipment, damage description, assessment, and applicable charge, and the record is linked to the member's loan.	Ensures equipment damage is documented and financial responsibility can be tracked.
NFR-001	Nonfunctional	The system shall display real-time physical equipment availability across neighborhood depot lockers.	**High**	**Pass:** After an equipment checkout or return, its availability status across the system is updated and displayed within **5 seconds**.	Accurate availability is essential because multiple members may attempt to borrow the same equipment.

NFR-002	Nonfunctional	The system shall restrict access to equipment-loan and damage-assessment information according to the user's assigned role.	**High**	**Pass:** Community members can access only their own loan information, while library custodians can access and modify relevant equipment and damage records. **Fail:** A member can access or modify another member's records or custodian-only data.	Protects member information and prevents unauthorized modification of library records.